Title: MQTT-based Messaging Layers: Toward Clock Synchronization Across Nodes in Smart Home Automation

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Abstract:

Clock synchronization across nodes remains a critical challenge for the Internet of Things, particularly in smart home automation. This paper proposes a framework built on MQTT-based messaging layers that explicitly accounts for over-the-air reliability. We evaluate the framework on a year-long smart-building trace and compare it against 7 baseline methods. Experimental results show that the proposed approach improves end-to-end delay by 30.5% while keeping end-to-end delay within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of the Internet of Things has changed how researchers and practitioners approach clock synchronization across nodes, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Despite considerable progress, existing solutions for clock synchronization across nodes tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in packet delivery ratio frequently come at the expense of detection rate, forcing designers to accept trade-offs that are poorly understood. This paper addresses these gaps by studying how MQTT-based messaging layers interact with energy-harvesting sensor nodes when applied to clock synchronization across nodes.

Our work is motivated by the growing demand for smart home automation, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of clock synchronization across nodes, a practical design built on MQTT-based messaging layers, and an empirical evaluation on a year-long smart-building trace. Understanding over-the-air reliability is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Subsequent work by Okafor et al. (2013) extended this line of research to smart home automation, although the evaluation was restricted to small-scale settings. A survey by Nakamura et al. (2010) categorised existing approaches to clock synchronization across nodes into three families and highlighted the lack of standard benchmarks. Researchers working on the Internet of Things have explored energy-harvesting sensor nodes as an alternative, with Fernandez et al. (2008) noting that the approach scales poorly beyond 8 components. Johansson et al. (2011) analysed over-the-air reliability and showed that purely heuristic solutions cannot guarantee consistent packet delivery ratio. More recently, Larsen et al. (2008) combined MQTT-based messaging layers with energy-harvesting sensor nodes, achieving a 36.8% reduction in overhead on a year-long smart-building trace.

Several studies have criticised the reliance on synthetic workloads, arguing that results on a year-long smart-building trace may not generalise to smart home automation. In contrast to these efforts, our work treats clock synchronization across nodes and over-the-air reliability jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to clock synchronization across nodes as a

persistent obstacle for smart home automation. Despite this body of literature, no prior study has systematically compared MQTT-based messaging layers and energy-harvesting sensor nodes under identical experimental conditions.

3. Methodology

We formalise clock synchronization across nodes as a constrained optimisation task in which the objective is to maximise end-to-end delay while keeping packet delivery ratio within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on MQTT-based messaging layers, and a validation layer that incorporates energy-harvesting sensor nodes. All experiments were conducted on a year-long smart-building trace, which provides 64 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 5 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies MQTT-based messaging layers to reduce the effect of over-the-air reliability, followed by a calibration step that stabilises packet delivery ratio. We compare against 7 baseline methods, including a conventional approach built on energy-harvesting sensor nodes and two recent variants reported in the literature. The proposed design improves end-to-end delay by 30.5% relative to the strongest baseline and reduces energy per transmitted bit by 11.0%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the energy-harvesting sensor nodes component reduces end-to-end delay by approximately 3.7%, confirming its contribution. A sensitivity analysis over 16 parameter settings shows that performance remains stable, with variation in energy per transmitted bit below 6.2%. Threats to validity include the limited diversity of a year-long smart-building trace and the possibility that deployment in smart home automation introduces conditions not captured in our setup.

4. Conclusion

This paper investigated clock synchronization across nodes through the lens of MQTT-based messaging layers, and demonstrated that a carefully designed pipeline can improve end-to-end delay by 30.5% on a year-long smart-building trace. The findings suggest that the Internet of Things will continue to benefit from closer integration of MQTT-based messaging layers and energy-harvesting sensor nodes, particularly in smart home automation. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of over-the-air reliability. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for clock synchronization across nodes.